

The Eighth Information Systems International Conference (ISICO 2025)

Design and development of an intuitive patient record management information system for a diagnostic clinic

Russel Acosta^a, Andrea Nicole Bautista^a, Maurell Ploya^a, Paul Bokingkito^{a,c,*}, Jerame Beloy^a, Jerina Ecleo^a, Rabby Lavilles^a and Lemuel Clark Velasco^{a,b,d}

^a*Department of Information Technology, College of Computer Studies, Mindanao State University-Iligan Institute of Technology, Iligan City, The Philippines*

^b*Department of Industrial and Information Management, College of Management, National Cheng Kung University, Tainan City, Taiwan*

^c*School of Science, Walailak University, Nakhon Si Thammarat, Thailand*

^d*Center for Computational Analytics and Modelling, Premiere Research Institute of Science and Mathematics, MSU-IIT, Iligan City, The Philippines*

Abstract

This study addresses inefficiencies in patient record management at a diagnostic clinic in Southern Philippines, where a decentralized manual system causes data redundancy and workflow issues. A Patient Record Management Information System (PRMIS) with AI-capable features was developed to centralize data, improve accessibility, and streamline operations. The Iterative Waterfall Model guided development through analysis, design, implementation, and testing. Business Process Reengineering was incorporated to enhance the current decentralized business process. Surveys and interviews across three clinic branches provided insights for system design and usability evaluation. Results showed high user satisfaction, with Branch 3 receiving the most favorable feedback. The intuitive PRMIS demonstrates strong potential to improve efficiency and enhance healthcare service delivery.

© 2026 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the Eighth Information Systems International Conference (ISICO 2025).

Keywords: patient record management information system; health information system; PRMIS; patient management system; diagnostic clinic; healthcare management systems

* Corresponding author.

E-mail address: pauljr.bokingkito@g.msuiit.edu.ph

1. Introduction

Efficient health information systems are vital for ensuring quality diagnostic care, especially in developing regions where manual processes remain prevalent [1], [2]. This study presents a case of system development and implementation in a real-world setting: a multi-specialty diagnostic clinic in Southern Philippines with 12 branches, where manual workflows have led to inefficiencies in patient record handling, result releasing, and inter-departmental coordination [3], [4].

This paper documents the development of an intuitive web-based Patient Record Management Information System (PRMIS) aimed at transforming fragmented, paper-based operations into a unified, automated digital process [5], [6], [7]. Unlike prior implementations that focus solely on technical development, this study frames the system design and implementation as a case study in organizational digital transformation. It highlights how the PRMIS was informed by actual workflows and user needs, reflecting the socio-technical complexity of healthcare information systems.

To understand and address workflow inefficiencies, the study employed Business Process Reengineering (BPR), which involves fundamentally redesigning processes to achieve significant improvements. Understanding the current ("as-is") and future ("to-be") states is key to pinpointing inefficiencies and guiding meaningful transformation. [8]. This supported a structured redesign of patient data management, queuing, and communication. The study aims to contribute to IS research by offering empirical insights into the implementation challenges, system usability, and design decisions within a complex healthcare environment.

Although the software development followed the Iterative Waterfall Model, this study's contribution lies in its contextual analysis, design choices grounded in actual processes, and its reflection on the outcomes of deploying IS solutions in a constrained healthcare setting [9]. It also integrates automation and AI-inspired features, such as queuing management and auto-notifications, not as standalone AI systems, but as pragmatic tools to reduce human workload and errors, adding operational intelligence to the system.

2. Methodology

This study employed a qualitative case study methodology to investigate the development and implementation of a Patient Record Management Information System (PRMIS) within a multi-branch diagnostic clinic in Southern Philippines. The case study approach was appropriate for exploring how an information system can address operational challenges in a real-world healthcare context, particularly those involving manual workflows, fragmented data handling, and coordination issues. Case study research is widely used in Information Systems to examine the interaction between technology, users, and organizational processes within their natural setting [1,2].

Data collection involved on-site observations, semi-structured interviews with clinic personnels including front desk staff, technologists, physicians, and administrators, and document analysis of internal protocols, patient forms, and spreadsheets across three representative branches. These activities informed the construction of both "As-Is" and "To-Be" process models using Business Process Model and Notation (BPMN 2.0) to visualize current inefficiencies and guide the definition of system requirements [8].

While the case study served as the overarching research methodology, the technical development of the PRMIS followed the Iterative Waterfall Model, a structured software engineering approach selected for its clarity and suitability for projects with well-scoped requirements [9]. The PRMIS was developed using this model (Fig. 3), progressing through requirement analysis, system design, implementation, and testing. Insights from the BPMN and stakeholder feedback informed key functionalities, which were formalized through use case diagrams and user interface prototypes. This distinction ensures that the system development lifecycle did not substitute for research methodology but rather complemented the case study framework by providing a structured process for building and deploying the IS artifact in context.

The system was implemented using Laravel (PHP) and ReactJS, and deployed in a local development environment using XAMPP. Usability was evaluated using the Post-Study System Usability Questionnaire (PSSUQ), administered to 11 clinic staff members who directly interacted with the system. While the sample size limits statistical generalizability, the evaluation provides valuable formative insights into the effectiveness of the design within the case context.

3. Results and Discussion

3.1. Business Process Identification and Redesign Results

The analysis of clinic operations revealed significant inefficiencies rooted in manual workflows, redundant documentation, and limited data access. Using data collected from interviews, observations, and document reviews across three branches, the researchers developed a BPMN 2.0 “As-Is” process model (Fig. 1) to map existing workflows in patient registration, laboratory requests, queuing, and result releasing. This model exposed process fragmentation and communication gaps, particularly in result retrieval and inter-departmental coordination.

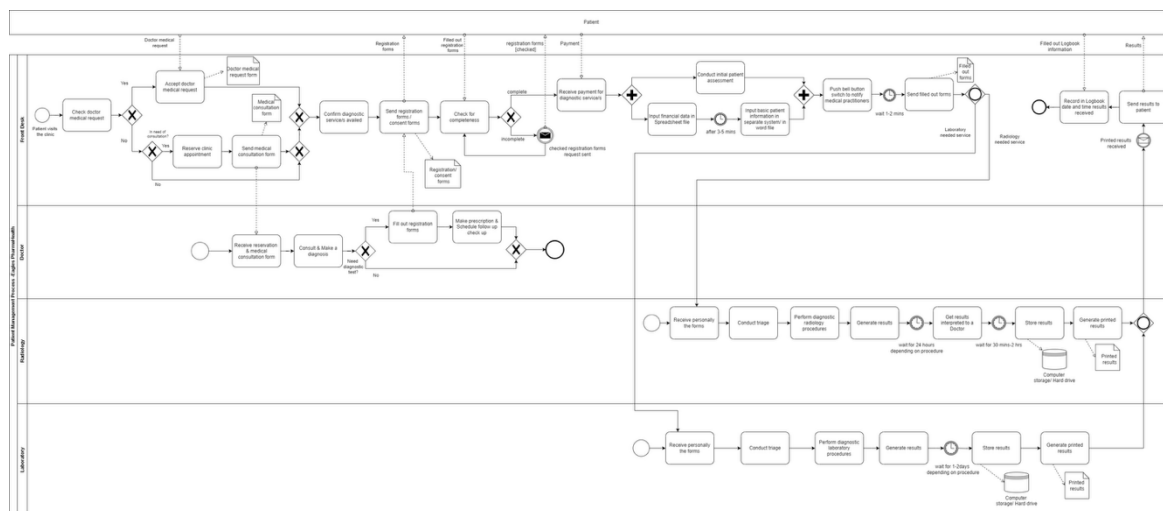


Fig. 1. As-is process model of the diagnostic clinic

To address these issues, a “To-Be” BPMN model (Fig. 2) was developed, illustrating a streamlined, system-enabled workflow guided by automation and centralized data management. The model proposed features such as intelligent queuing, real-time record access, and automated result notifications—core functionalities that shaped the requirements of the Patient Record Management Information System (PRMIS). This alignment between BPMN insights and system design ensured that the PRMIS addressed practical clinic needs while contributing to broader goals of process reengineering in healthcare information systems.

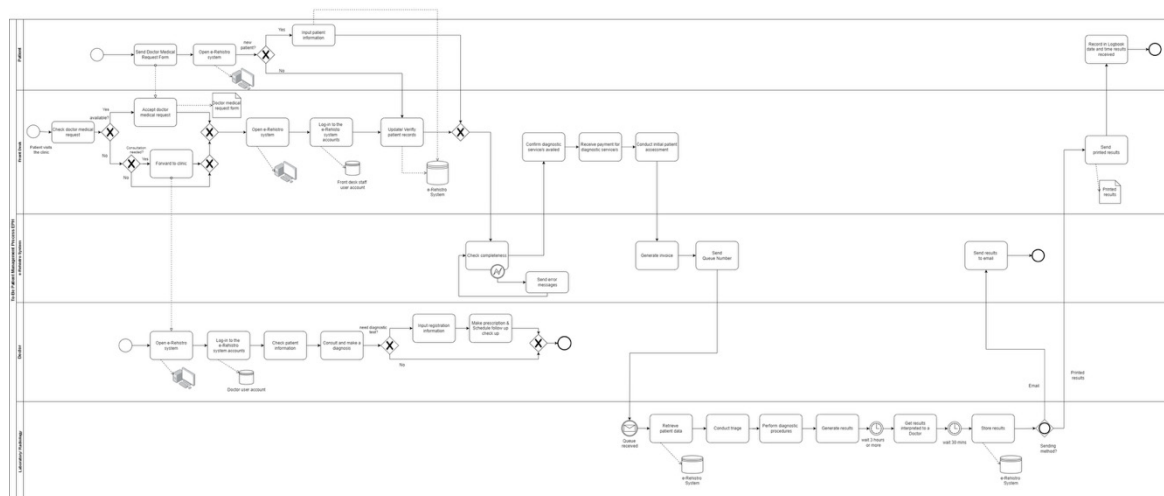


Fig. 2. To-be process model of the diagnostic clinic

3.2. System Implementation Results

The use case diagram (Fig. 3) outlined six primary functions: patient registration, laboratory services, transactions, queuing, result releasing, and consultations—mapped against specific system actors. This ensured that users, including front desk staff, technologists, and physicians, interacted with the system in ways that reflected their roles and responsibilities, supporting usability and minimal workflow disruption.

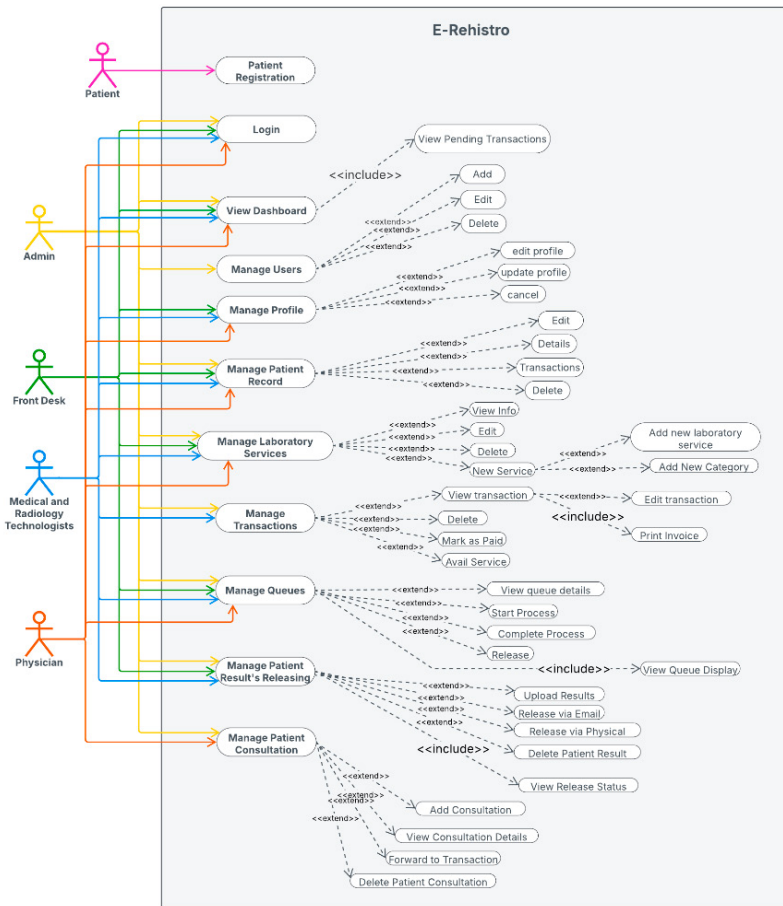


Fig. 3. Use case diagram of the patient record management information system

The system architecture (Fig. 4) demonstrates the modular structure of PRMIS, integrating a web-based platform with core modules and services, including user management, patient records, consultations, queue handling, and result delivery. It also incorporates role-based authentication and connects to essential services such as data storage and email. Rather than serving as a static infrastructure view, the architecture reflects how secure, centralized access enables operational continuity and responsive care delivery across branches.

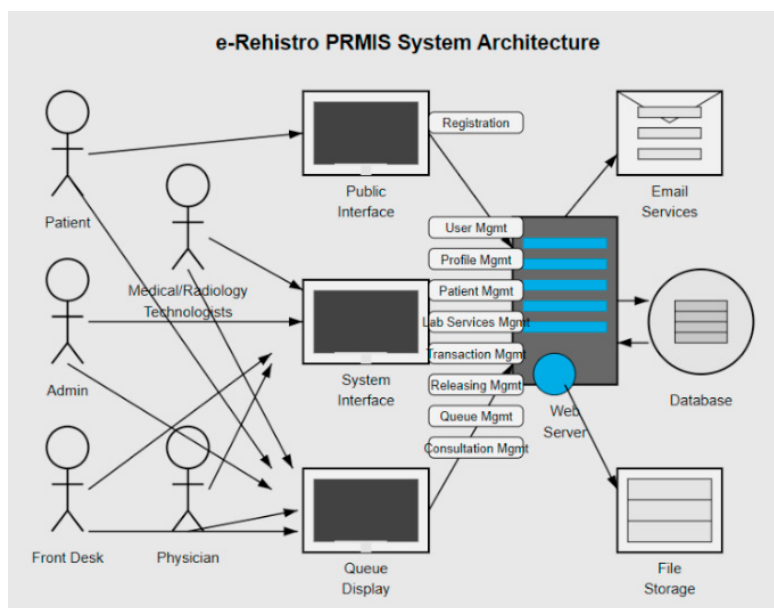


Fig. 4. System architecture of the patient record management information system

The final system was built using Laravel (PHP) and ReactJS, deployed via XAMPP. Module development was iterative, incorporating feedback from clinical staff to ensure practical utility and ease of adoption. The results of this case demonstrate that process-driven design and stakeholder engagement significantly enhance the relevance, usability, and impact of information systems in healthcare settings (see system figures below).

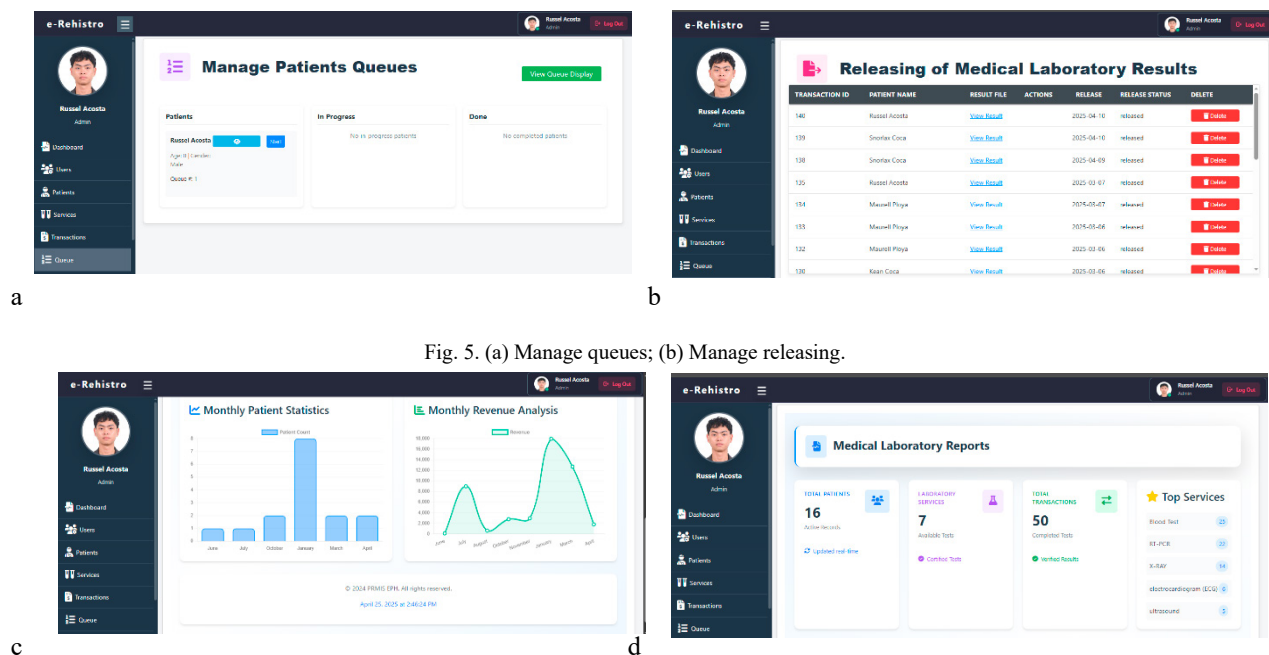


Fig. 5. (c) Reporting module (d) Dashboard of the PRMIS.

3.3. Security and Usability Evaluation Results

As shown in Fig. 6, the security testing of the PRMIS included rigorous evaluation through software composition analysis using SOOS, which confirmed no reported vulnerabilities or violations of government security standards. However, beyond dependency-level validation, the system was also designed with multiple layers of application-level security to address key healthcare-specific threats such as data breaches, identity spoofing, and unauthorized access.

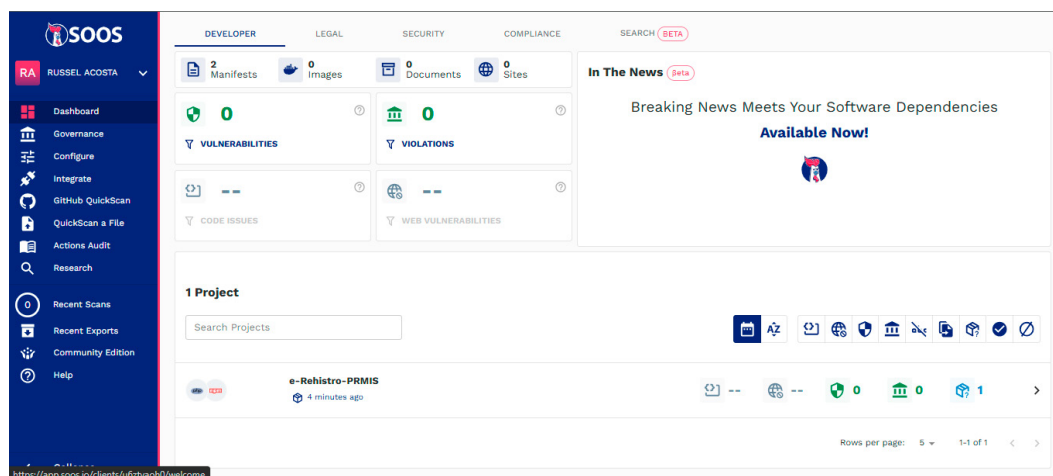


Fig. 6. SOOS analysis for the intuitive PRMIS

To protect against Cross-Site Request Forgery (CSRF) attacks, PRMIS implements Laravel's built-in CSRF protection mechanism. All form submissions include CSRF tokens that are validated server-side, preventing malicious requests from unauthorized sources. This is critical in safeguarding authenticated user sessions, particularly in administrative and clinical interfaces where sensitive actions are performed. For password storage, PRMIS uses bcrypt, a well-established hashing algorithm known for its adaptive strength. This ensures that even if password hashes are exposed, recovering the original passwords remains computationally infeasible.

Furthermore, to protect confidential patient data at rest, the system uses AES-256 encryption, an industry-standard symmetric encryption algorithm recognized for its robustness and compliance with HIPAA and other healthcare data protection regulations. AES-256 is applied to fields such as patient medical histories and test results, ensuring that sensitive data is unreadable even if storage media are compromised.

These security measures are embedded into the system's core architecture, minimizing reliance on external tools and manual oversight. They address critical elements of the CIA triad (Confidentiality, Integrity, Availability) and align with secure-by-design principles [10]. By combining automated analysis (via SOOS) with application-level safeguards (CSRF, bcrypt, AES-256), PRMIS delivers a multi-faceted security approach appropriate for real-world deployment in healthcare environments.

To assess system usability, the Post-Study System Usability Questionnaire (PSSUQ) was administered to 11 clinic personnel across three branches. While the use of PSSUQ a widely accepted instrument ensured methodological rigor, the small sample size presents limitations in terms of generalizability, as it primarily offers formative rather than statistically robust insights. Nevertheless, as shown in Fig. 7 the evaluation served as an initial validation of system intuitiveness from the perspective of core users actively engaged in the clinic's daily operations. The results revealed high user satisfaction, with System Usefulness receiving the most favorable mean score (1.58), followed by Interface Quality (1.69) and Information Quality (1.90). These findings suggest that the system supports users in completing tasks efficiently and comfortably.

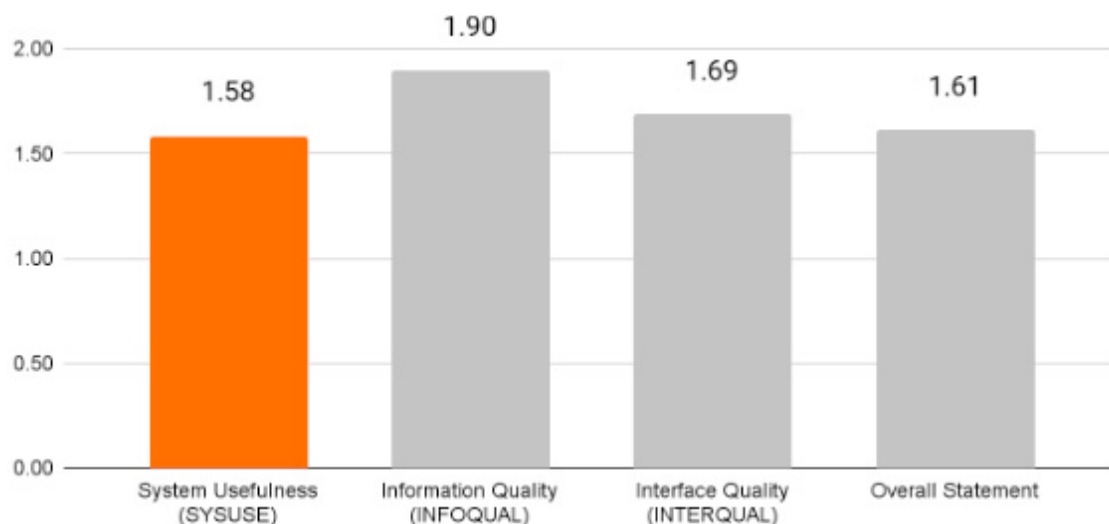


Fig. 7. PSSUQ score for the intuitive PRMIS

More importantly, the favorable feedback aligns with key principles from IS usability frameworks, particularly ease of learning, effectiveness, and satisfaction dimensions emphasized in Nielsen's usability heuristics and the ISO 9241-11 standard. In this case, the high ratings on system usefulness reflect strong alignment with task-support effectiveness, while ease of learning was observed in the minimal user onboarding required across branches. Branch 3, which had better infrastructure and administrative support, showed the highest usability ratings, highlighting how organizational readiness can affect perceived usability. Conversely, lower scores from Branch 2 suggest that contextual constraints such as limited training and staff capacity impact the system's perceived intuitiveness. While broader testing is needed for general claims, this early-stage evaluation provides empirical grounding for the system's design decisions and indicates that process-aligned development can lead to meaningful improvements in user experience within constrained clinical settings.

4. Conclusion

The development and implementation of the Patient Record Management Information System (PRMIS) in a multi-branch diagnostic clinic demonstrates how health information systems can address operational inefficiencies in resource-constrained environments. By digitizing fragmented, paper-based workflows, the system improved coordination, reduced delays, and enhanced data accessibility across departments. The use of Business Process Model and Notation (BPMN) to analyze and redesign existing workflows ensured that the system's functionalities were grounded in real operational needs. Usability evaluation using PSSUQ revealed high satisfaction among clinic staff, affirming the system's effectiveness and intuitiveness within its deployment context.

Beyond its practical impact, this study contributes to the Information Systems (IS) discipline by showcasing how a qualitative case study methodology can generate actionable insights into IS implementation. The research emphasizes the importance of aligning system design with actual workflows and organizational readiness—core principles in socio-technical systems theory. Moreover, the integration of automation features and process-driven design decisions underscores how local innovations in constrained settings can yield transferable knowledge for broader IS applications.

The study also highlights how user-centered development, informed by real-time feedback and contextual analysis, supports usability and system adoption. Through its use of established usability frameworks (e.g., ISO 9241-11,

Nielsen's heuristics) and reflection on design outcomes, the paper offers a theoretically informed perspective on the challenges and enablers of IS adoption in healthcare. These findings contribute to the ongoing discourse on digital transformation, particularly in low-resource contexts, and invite further exploration of how grounded, iterative development approaches can extend IS knowledge and practice in healthcare settings.

Appendix

Complete figures of the PRMIS are found here: <https://github.com/Namdosan12-ui/Design-and-Development-of-an-Intuitive-PRMIS-for-a-Diagnostic-Clinic>

References

- [1] Zhou, L., Soran, C., Jenter, C. A., Jiang, J. P., & Kaushal, R. (2009) "The Relationship between Electronic Health Record Use and Quality of Care over Time." *Journal of the American Medical Informatics Association* 16 (4): 457–464. <https://doi.org/10.1197/jamia.M3128>.
- [2] Bala, G. C., Islam, M., & Mamun, S. A. (n.d.) "Patients Management System using Fuzzy Logic: An Easier and Dynamic Appointment System."
- [3] Nhavoto, J. A., Grönlund, Å., & Chaquilla, W. P. (2015) "SMSaúde: Design, Development, and Implementation of a Remote/Mobile Patient Management System to Improve Retention in Care for HIV/AIDS and Tuberculosis Patients." *JMIR mHealth and uHealth* 3 (1): e26. <https://doi.org/10.2196/mhealth.3854>.
- [4] Pambudi, I. T., Hayasaka, T., Tsubota, K., Wada, S., & Yamaguchi, T. (2004) "Patient Record Information System (PaRIS) for Primary Health Care Centers in Indonesia." *Technology and Health Care* 12 (4): 347–357. <https://doi.org/10.3233/THC-2004-12407>.
- [5] Emmanuel, E. (2009) "An Automated System for Patient Record Management (A Case Study of St Francis Hospital Nsambya)." [Online]. Available: <http://hdl.handle.net/10570/443>.
- [6] Rajput, S., Vadivel, D. S., & Shetty, S. D. (2010) "Design and Security Analysis of Web Application-Based and Web Services-Based Patient Management System (PMS)."
- [7] Bouayad, L., Ialynytchev, A., & Padmanabhan, B. (2017) "Patient Health Record Systems Scope and Functionalities: Literature Review and Future Directions." *Journal of Medical Internet Research* 19 (11): e388. <https://doi.org/10.2196/jmir.8073>.
- [8] Aagesen, G., & Krogstie, J. (2015) "BPMN 2.0 for Modeling Business Processes." In Vom Brocke, J., & Rosemann, M. (eds), *Handbook on Business Process Management 1*. Berlin, Heidelberg: Springer Berlin Heidelberg, 219–250. https://doi.org/10.1007/978-3-642-45100-3_10.
- [9] Adenowo, A. A. A., & Adenowo, B. A. (2013) "Software Engineering Methodologies: A Review of the Waterfall Model and Object-Oriented Approach." *International Journal of Engineering Research and Applications* 4 (7).
- [10] Osazuwa, O. M. C. (2023) "Confidentiality, Integrity, and Availability in Network Systems: A Review of Related Literature." *International Journal of Computer Applications* 8 (12).